Increasing and decreasing functions

1

a Decreasing

b Increasing

2

a 1

b $(1, \infty)$

c $(-\infty, 1)$

3 $(-\infty, 2)$

4

a

i $(-\infty, -2) \cup (2, \infty)$

ii $(-2, 2)$

b

i $(-1, 0) \cup (1, \infty)$

ii $(-\infty, -1) \cup (0, 1)$

5

a $(0, 0)$

b $(0, 0)$

c Increasing

6

a $(-1, 0)$

b Decrease

c Increase

Zeros of functions

7

a

i $(4, 0), (-2, 0)$

ii $4, -2$

b

i $(2, 0), (-4, 0)$

ii $2, -4$

c

i $(3, 0)$

ii 3

8

a -3

b $3, -1$

c $5, 3$

End behavior

9

a $(-4,0),(0,0)$ b $(0,0)$ c ∞ d ∞

10

a The function does not have x-intercepts. b $(0,1)$

c ∞

11

a

i $(6,0),(0,0)$ ii $(0,0)$ iii Increasing iv Increasing

b

i $(-6,0),(6,0)$ ii $(0,36)$ iii Decreasing iv Decreasing

c

i $(-1,0),(1,0)$ ii $(0,1)$ iii Decreasing iv Increasing

d

i $(3,0)$ ii $(0,7)$ iii Decreasing iv Increasing

e

i $(-4,0),(0,0),(4,0)$ ii $(0,0)$

iii Increasing iv Increasing

f

i $(-2,0),(-1,0),(3,0),(4,0)$ ii $(0,4)$

iii Increasing iv Increasing

g

i $(-4,0),(3,0),(1,0)$ ii $(0,-6)$

iii Decreasing iv Increasing

h

i The function does not have x-intercepts.

ii $(0,1)$

iii Increasing

iv Decreasing

i

i $(0,0),(4,0),(0,3)$ ii $(0,0)$

iii Decreasing iv Decreasing

j

i $(0, 0)$ ii $(0, 0)$ iii Increasing iv Decreasing

Asymptotes

12

a 0 b $y = 0$ c True

13

a 1 b $y = 1$ c True

14

a

i $-\infty$ ii ∞ iii 0 iv 0

b $x = 0$

c $y = 0$

15

a

i ∞ ii $-\infty$ iii 4 iv 4

b $x = 5$

c $y = 4$

16

a

i ∞ ii ∞ iii 0 iv 0

b $x = 1$

c $y = 0$

17

a

i 1 ii 1

b $y = 1$ c $(-\infty, \infty)$